

1.1. INTRODUCTION

Cybersafe Community Platform is a secure, role-based web application designed to connect cybersecurity enthusiasts, professionals, and individuals seeking expert guidance in the field. Built with **React.js** for the frontend, **Node.js** for the backend API, and **MySQL** for database management, the platform fosters a collaborative environment where users can share knowledge, resolve technical challenges, and stay updated on cybersecurity trends. Seekers can post questions, browse verified Cyber Experts based on qualifications (e.g., certifications like CISSP or CEH), and receive tailored solutions. Cyber Experts, approved exclusively by Admins for credibility, provide detailed answers, share practical insights, and showcase their expertise through structured profiles. Admins oversee the platform's integrity by managing user roles, approving/rejecting experts, and monitoring feedback to ensure compliance with security protocols.

MODULES :

Admin

- Can login and log out to the system
- Can add and remove cyber experts
- Can ensure community guidelines Can manage overall platform

Cyber experts

- Can sign in to the system
- Can view queries and answer to the user queries
- Can discuss among other cyber experts

Seeker

- Can sign in and signup to the system
- Can ask questions, seek advice and participate in discussions
- Can choose cyber experts of our own for enquiring

2. SYSTEM ANALYSIS

2.1. INTRODUCTION TO SYSTEM ANALYSIS

System analysis is the process of evaluating a business or operational framework with the goal of enhancing efficiency through optimized procedures and technological solutions. In the context of the Cybersafe Community Platform, system analysis plays a pivotal role in scrutinizing the existing methods of cybersecurity knowledge exchange and transforming them into a secure, scalable, and user-centric digital ecosystem.

The current landscape for cybersecurity guidance often relies on public forums, informal networks, or fragmented tools. These systems lack structured verification of experts, centralized oversight, and role-based access control. Manual processes dominate, leading to inefficiencies such as delayed responses, unverified advice, and challenges in tracking credible solutions.

Through rigorous system analysis, the Cybersafe Community Platform identifies these shortcomings and proposes a modern web application built with React.js (frontend), Node.js (backend), and MySQL (database). The platform introduces three distinct user modules:

1. Seekers: Users seeking cybersecurity guidance.
2. Cyber Experts: Verified professionals providing solutions.
3. Administrators (Admins): Overseers ensuring platform integrity and security.

Current System Issues:

- Reliance on unregulated forums with unvetted advice.
- No structured workflows for query resolution or expert validation.
- Lack of accountability, real-time updates, and secure communication.

By conducting a comprehensive system analysis, the Cybersafe Community Platform ensures the final solution addresses critical pain points, reduces reliance on unreliable resources, and fosters a trusted environment for cybersecurity collaboration.

2.2. EXISTING SYSTEM

In the current cybersecurity landscape, individuals and organizations seeking guidance often rely on fragmented, informal channels such as public forums, social media groups, or direct consultations. These methods present significant limitations:

1. Manual and Unregulated Process:

- Seekers must scour multiple platforms to find answers, often encountering unverified or outdated advice.
- No centralized system exists to validate the expertise of those providing solutions, increasing the risk of misinformation.

2. Time-Consuming and Risky:

- Critical cybersecurity queries may go unanswered for days, leaving vulnerabilities unaddressed.
- Users depend on the availability of knowledgeable individuals, with no guarantee of timely or accurate responses.

3. Lack of Role-Based Security:

- Anyone can pose as an expert, leading to potential misuse, unauthorized advice, or malicious guidance.
- No structured permissions or accountability mechanisms exist to distinguish between Seekers, verified Experts, and Admins.

2.3. PROPOSED SYSTEM

The **Cybersafe Community Platform** is a web-based solution designed to address the inefficiencies of traditional cybersecurity knowledge-sharing methods. Built with **React.js** (frontend), **Node.js** (backend), and **MySQL** (database), the platform provides a centralized, secure hub for Seekers to connect with vetted Cyber Experts, ensuring reliable and timely resolution of cybersecurity queries.

Key Features and Advantages:

1. Centralized Cybersecurity Hub:

- All questions, answers, and resources are stored in a secure, searchable database.
- Users access verified solutions instantly, eliminating reliance on fragmented forums.

2. Secure User Authentication:

- JWT token-based login ensures only registered users can interact with the platform.

- Role-based access (Seeker, Cyber Expert, Admin) restricts unauthorized actions.

3. **Role-Based Access Control:**

- **Seekers:** Post questions, rate answers, and browse expert profiles.
- **Cyber Experts:** Submit answers, showcase certifications (e.g., CISSP, CEH), and manage response history.
- **Admins:** Approve/reject experts, delete malicious users, and monitor platform activity.

4. **Efficient Knowledge Sharing:**

- Experts upload detailed responses with code snippets, logs, or screenshots (stored securely).
- Seekers download solutions or mark resolved queries for future reference.

5. **Structured Data Management:**

- MySQL organizes user profiles, questions, answers, and certifications for fast retrieval.
- Automated backups and encryption ensure data integrity and security.

6. **User-Friendly Interface:**

- React.js delivers an intuitive dashboard with real-time notifications and filters (e.g., filter experts by certification).
- Clean design simplifies navigation for users of all technical levels.

7. **Time-Saving and Secure:**

- Reduces time spent searching for credible advice through a centralized, vetted platform.
- SSL encryption and input sanitization protect against data breaches and attacks.

8. **Admin Oversight:**

- Admins validate expert credentials, remove harmful content, and generate usage reports.
- Audit logs track user activity to ensure compliance with security policies.

2.4. **USER CHARACTERISTICS**

The platform operates with three user modules:

1. **Admin**

- Oversees platform security, user compliance, and content integrity.
- Approves/rejects **Cyber Expert** registrations after credential verification.

- Monitors all questions, answers, and user interactions for policy violations.
- Views audit logs, manages user bans, and generates activity reports.

2. Cyber Experts

- Submit detailed, vetted answers to cybersecurity questions (e.g., code fixes, threat analysis).
- Showcase certifications (e.g., CISSP, CEH) and response accuracy metrics.
- Update profiles with new qualifications and manage answer history.
- Flag inappropriate content or malicious queries for admin review.

3. Seekers

- Register and post cybersecurity questions with tags (e.g., #Ransomware, #NetworkSecurity).
- Browse and filter experts by certifications, ratings, or response times.
- Download solutions, rate answers for accuracy, and mark queries as resolved.
- View expert profiles and historical answers for informed decision-making.